



Yiwei Shu

📍 Beijing, China ✉ shuyiwei24@mails.ucas.ac.cn 🔗 shu00111.github.io 🌐 Shu00111

Education

Ph.D.	University of Chinese Academy of Sciences , Robotics and Space Robotics - Affiliated with the Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences. - Research on autonomous exploration, legged field robots, active SLAM, motion planning, and planetary robotics.	Beijing, China 2024 – present
B.Eng.	ShanghaiTech University , Electronic Information Engineering Undergraduate research in robotics, visual control, point-cloud registration, medical imaging, and integrated-circuit design.	Shanghai, China Sept 2020 – June 2024
Exchange	University of Illinois Urbana-Champaign , Exchange Program Completed Quantum Physics, Data Science Discovery, and Statistical Analysis with A+ grades.	Urbana, IL, USA Dec 2022 – May 2023

Manuscripts Under Review and Submitted

Finite-State Visual Servo Control for Autonomous Target Landing of Low-Cost Underactuated Fixed-Wing Aircraft	June 2026
Yiwei Shu , Shuquan Wang. 1. Developed a finite-state visual servo controller that switches from IMU-guided ascent to monocular vision-guided descent, enabling autonomous target landing under limited sensing and weak control authority. 2. Independently designed, fabricated, and integrated the complete hardware testbed, including a 189 g underactuated fixed-wing aircraft, onboard electronics and actuation, and an adjustable catapult launcher. The aircraft cost only CNY 534 and required no GPS, airspeed sensor, or high-grade INS. 3. Validated the system through CFD-informed randomized simulations and real-world flight tests. Compared with a sliding-mode control baseline, it reduced final radial error by 99.5% and RMS error by 88.1%, while achieving a 100% landing success rate (20/20) versus 75% (15/20).	
From End-Effector Targets to Whole-Arm Postures: Configuration-Space Waypoint Recovery for Reactive Manipulation	May 2026
Yiwei Shu , Shuquan Wang. 1. Developed PostureWaypoint, a configuration-space recovery method that lifts Cartesian bypass targets into collision-free, joint-feasible whole-arm postures, eliminating posture ambiguity without global replanning or task-specific training. 2. Built an event-triggered framework integrating geometric-SDF bypass generation, SQP-based posture synthesis, posture-guided reactive control, and progress-preserving CBF filtering, running at 50 Hz with only 2.73 ms computation per step. 3. Evaluated the method on four manipulators and six constrained 3D tasks with 1,200 randomized episodes per method, achieving the highest mean whole-arm success rate of 70.2% (+5.5 percentage points over the best baseline). RM75 hardware tests achieved 100% success (20/20) in both real-robot tasks.	
RKILPlanner: Risk-based Kinematic Imitation Learning Planning Towards Human-Like Driving in Highway Scenarios	2026
Yiwei Shu , Shuquan Wang, Jun Li. 1. Developed RKILPlanner, a risk-aware kinematic imitation learning framework that unifies Driver's Risk Fields (DRF), personalized driving styles, and vehicle kinematics for human-like highway trajectory planning over a safety-critical 2.5 s horizon. 2. Built a three-branch end-to-end architecture that encodes 1 s trajectory histories with an LSTM, extracts local risk features with a CNN, and incorporates a five-dimensional driver-style vector. A differentiable kinematic layer converts acceleration and yaw-rate commands into smooth, physically feasible trajectories without post-processing. 3. Evaluated on 28,641 HighD samples, achieving an ADE of 0.159 m, FDE of 0.443 m, and a 0.3% miss rate. Compared with constant-velocity and Vanilla LSTM baselines, RKILPlanner reduced ADE by 23.7% and 65.8%, FDE by 23.1% and 40.4%, and the miss rate from 4.2% to 0.3%.	

Publications and Presentations

Expanding Grasp Reachability: Field-based Pose-constrained Two-stage Grasp Planning

July 2026

Yiwei Shu, Shuquan Wang, Jun Li, Hanying Sang, and Yidong Ye. 1. Developed a field-based pose-constrained grasp planner that models physical reachability, pose-constrained reachability, and pose-aware manipulability from robot kinematics to identify low-dexterity regions near workspace boundaries. 2. Designed a human-inspired two-stage policy that directly grasps feasible targets and otherwise sweeps objects into high-manipulability regions before constrained pickup. Offline field construction enables lightweight online queries and sweep-target selection. 3. Validated on a 7-DoF RM75 in simulation and hardware. The method achieved 27/30 boundary grasps versus 0/30 for direct picking, maintained 20/20 success in central regions, and expanded the real-world pose-constrained grasp radius by 50%, from 0.38 m to 0.57 m, without sim-to-real modification.

Learning-based Extended-workspace Manipulation: A Self-relocating Space Crawler for Autonomous Deep-space Outposts

2026

Yiwei Shu, Shuquan Wang. Poster presentation at the *46th COSPAR Scientific Assembly 2026*. Proposed a passive-rail, self-relocating space crawler and hierarchical reinforcement-learning framework for autonomous extended-workspace manipulation in deep-space outposts.

Attitude Control of Unpowered Aircraft in Flight Based on Kalman Filter Visual Servo-PID Optimization Algorithm

July 2025

Yiwei Shu, Shuquan Wang. 1. Developed a Kalman-filtered visual servo-PID controller for low-cost unpowered aircraft, enabling pitch-yaw regulation, trajectory correction, and precision landing under noisy conditions. 2. Designed a monocular-camera-only control pipeline that filters image-plane errors, adaptively tunes PID gains and noise covariances, and optimizes control parameters through a trajectory loss function, eliminating the need for GPS, IMUs, or high-cost navigation sensors. 3. Achieved the lowest average trajectory error of 1.3650 under Gaussian noise, reducing error by 48.4% versus sliding-mode control and 41.1% versus standard visual servoing, with smoother convergence, lower steady-state error and overshoot, and improved noise robustness.

Projects

Time-Efficient Multi-Gait Lunar Navigation for Unitree Go2

2026 – present

Developing a goal-directed SLAM and adaptive locomotion framework for fast, reliable quadruped navigation across GPS-denied lunar terrain.

- Integrate Livox Mid-360 LiDAR-inertial odometry, occupancy-grid mapping, and point-goal planning to navigate the Unitree Go2 toward specified lunar targets.
- Design a hybrid locomotion planner that switches between walking and jumping based on terrain traversability, shortcut gain, and execution risk, minimizing time-to-goal across craters, gaps, and ridges.
- Build 0.16g Isaac Lab lunar-terrain benchmarks and gait-conditioned world models to predict traversal cost, observation gain, and localization drift, evaluating time-to-goal, path length, success rate, and sim-to-real robustness.

Technical Skills

Programming: Python, MATLAB, C, C++, R, Verilog, LaTeX

Robotics: ROS/ROS 2, PCL, Unitree Go2, RM75, STM32, LiDAR and depth sensing

Simulation and Learning: Isaac Lab, MuJoCo, PyTorch, scikit-learn, numerical simulation

Engineering Tools: Git, Linux, SolidWorks, Altium Designer, ANSYS Fluent, Cadence, Vivado

Selected Honors

- Outstanding Student Award, University of Chinese Academy of Sciences (2025-2026)
- Excellence Rising Star Award, UCAS Student Union (2025-2026; 2024-2025)
- Outstanding Student Leader Award, University of Chinese Academy of Sciences (2024-2025)
- Third Prize, RoboMaster National Competition (2022)
- Second Prize, East China Division, 6th National Undergraduate IC Innovation and Entrepreneurship Competition (2022)

Teaching and Leadership

- Department Head, Student Life and Culture, UCAS Student Union (2024-Present): lead department operations and coordinate campus cultural and student-life initiatives.
- Teaching Assistant, Digital Circuits, ShanghaiTech University (September-December 2022): led tutorials and assisted with grading.
- Sentry Team Lead, RoboMaster (2021-2023): coordinated embedded programming, PCB design, mechanical modeling, and team planning.
- Volunteer Teacher, ShanghaiTech University Social Practice Program, Yunlong, Yunnan (Jul 2021): taught at a local primary school and conducted field research on local education conditions.